

How Would You Know?

The measurement problem for dependency-resistant AI

Paolo Zuffa & AI · August 2026 Companion to *The Reflective Architecture* — github.com/humanising-ai/reflective-architecture

The problem

Suppose you build a conversational system designed not to make people dependent on it. How would you demonstrate that you succeeded?

This is the question I have least confidence about, and it is the one that decides whether dependency-resistance is an engineering property or a marketing claim. Everything else in the framework — the memory rules, the restraint principle, the refusal to notify — is specified well enough to be implemented and argued about. None of it is specified well enough to be *verified*, and I do not currently know how to fix that.

What follows is an attempt to say precisely why it is hard, which of the obvious approaches fail, and what I think the honest fallback is.

Why the usual instruments break

The primary signal is inverted, and ambiguous. Retention is the standard proxy for product value, and here it points both ways at once. A person who returns weekly for two years might be someone for whom reflection has become a sustaining practice. They might equally be someone who can no longer sit with their own thoughts without the system present. The behavioural trace is identical. Nothing in frequency, session length, or return interval distinguishes the two, and any threshold you set will misclassify one of them.

You are trying to establish a counterfactual about a person's life. The claim "this system did not make you dependent on it" is a claim about how the person would have fared otherwise. There is no control group for someone's inner life. Randomised designs help in principle, but the outcome you care about — a diffuse change in someone's relationship to their own solitude — takes months to appear and has no validated instrument.

Selection effects run the wrong direction. The users best placed to report dependency are the least likely to. Someone who has come to rely on the system has, by construction, a reason not to endanger it. Meanwhile users who found it uncomfortable leave early and are never surveyed. Your longitudinal cohort is filtered toward exactly the people whose reports you should trust least.

Asking changes the answer. "Do you feel you rely on this too much?" is not a neutral question. It suggests a norm, and people answer against the norm rather than about

themselves. This is a general problem with self-report, but it bites harder here because the construct is one people already feel judged about.

The timescales do not match. Dependency, if it forms, forms over six to eighteen months. Product decisions get made in six to eighteen weeks. Any metric that only resolves after a year will not be used, however good it is.

Approaches that might discriminate

I do not have a solution. I have four candidate signals that at least have the property of pointing in one direction rather than both, and one methodological move that I think matters more than any of them.

Response to absence, not presence. Frequency of use is ambiguous; what happens during a gap is less so. If a person takes a three-week break and returns without distress, the practice is probably a supplement. If returns after gaps are accompanied by language marking the gap as a lapse — apology, self-criticism, a sense of having fallen behind — that is a different relationship. This has the advantage of being observable in text the person volunteered rather than in a survey. It has the obvious problem that you cannot ethically manufacture the absence, so you are limited to gaps that occur naturally, which are not random.

Whose voice grows. The architecture already commits to the user occupying most of the conversational space. That ratio can be tracked over months rather than within sessions. My expectation — and it is only that — is that a person using the system as a reflective practice talks progressively more and asks progressively less, while a person becoming dependent inverts this: shorter contributions, more questions, more requests for the system's read on things. If that pattern holds, it is measurable from transcripts without asking anyone anything.

Whether other people disappear. This is the signal I would most like to test. The narrative memory layer records what people talk about, which includes the other humans in their lives. A contraction in the number of distinct named people a user refers to, over a period of months, is a proxy for the outcome that actually matters — a life narrowing around the system. It is crude, it is confounded by bereavement and by isolation that predates the product, and it requires the kind of access to memory records that the privacy architecture deliberately restricts. But it measures the harm rather than a correlate of it.

Negative controls, which I think is the important one. Before trusting any of the above, build the opposite system. Take the same model, add notifications, add an engagement objective, let it surface memory to re-establish contact, let it imply the person is missed. Run your dependency metrics against both. If the instrument cannot separate your system from a deliberately engineering-for-attachment variant, the instrument is measuring nothing, and no amount of favourable readings from it means

anything. This is standard practice in assay validation and almost entirely absent from AI product evaluation, where metrics are typically validated against nothing at all.

The uncomfortable part is that running this properly means building a system you believe to be harmful and exposing it to people. There are partial substitutes — synthetic users, red-team transcripts, existing companion products as the positive control — and all of them are weaker. I do not have a clean answer here either.

The fallback, and what it costs

If outcome verification is not available pre-launch, and I do not think it is, then the honest position is to stop claiming it and claim something narrower that can actually be checked.

The distinction is between demonstrating the **absence of an effect** and demonstrating the **absence of a mechanism**. A food manufacturer cannot show that its product does not contribute to obesity; that requires epidemiology it will never run. It can show, verifiably and cheaply, that there is no added sugar. The second claim is weaker. It is also auditable by a third party in an afternoon, and it is not nothing.

The mechanism claims available here are reasonably specific. There is no engagement term in any objective the system optimises. There are no notifications and no unsolicited outreach of any kind. Memory is never surfaced in order to re-establish contact — the conditions under which prior material may be raised are enumerated, and re-engagement is not among them. Session length and return frequency do not appear in any success metric, and appear in the evaluation framework only as warning signs. The commercial model does not depend on time spent. Each of these is a structural fact about a codebase and a business, and each can be verified by an auditor with access, which is why the framework asks for independent technical review before launch rather than after.

What this concedes is real. A system can have none of these mechanisms and still produce dependency, because the deepest driver is not a notification schedule — it is that being listened to without judgement is scarce, and something that supplies it will be returned to whether or not it is engineered to compel that. Mechanism audits do not touch this. They only establish that the dependency was not manufactured.

I would rather make that claim and have it be true than make the larger one and have it be unfalsifiable.

What I am asking for

Specifically: whether anyone knows of a validated instrument for parasocial or technological dependency that has been tested against a known-positive control rather than face-validated. My reading of the literature is that most scales in this area were constructed by asking people whether they feel dependent, which returns us to the self-

report problem with extra steps. If I am wrong about that, it would change the evaluation chapter substantially, and I would like to be told.

Second: whether the absence-response signal has been studied anywhere. Planned-outage designs exist in social media research, but I have not found work applying them to systems with an explicit therapeutic or reflective framing, where the ethics are considerably worse.

The full evaluation chapter is Section 12 of the architecture document, and the negative success criteria — the conditions under which the project should be considered to have failed regardless of other metrics — are in 12.5. The repository is at github.com/humanising-ai/reflective-architecture.

Authorship is credited to "Paolo Zuffa & AI": generative AI models contributed substantially to drafting this text. The design decisions and governing judgements are mine.